

Graph Theory and Applications

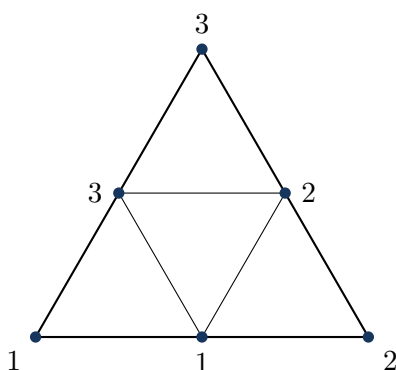
Session 2 — From counting triangles to finding fixed points

Master 1 APE

1 Must a three-coloured triangle contain all three colours?

We divide a triangle into smaller triangles and label their vertices with 1, 2, 3. The three corners receive different labels, and along each side we use only its two endpoint labels. Interior vertices may receive any of the three labels.

Can we choose the labels so that *no* small triangle has all three labels? Try to answer from the example below before attempting a proof.



Part I — Counting changes along a boundary

Before studying the two-dimensional picture, consider a finite sequence $x_0, x_1, \dots, x_r$, where $r \geq 0$ and every x_j is either 1 or 2. A *change* is an index $j \in \{0, \dots, r-1\}$ such that $x_j \neq x_{j+1}$. Let $c(x_0, \dots, x_r)$ be the number of changes; for $r = 0$, this number is zero.

- 1.1. Compute the number of changes in $(1, 2, 2, 1, 2, 2)$. Check all eight sequences with three entries. What relationship do you observe between the first and last entries and the parity of the number of changes?
- 1.2. Formulate this relationship as a precise proposition. Prove it by induction on r . Start with $r = 0$. In the induction step, distinguish whether the new final entry agrees with the previous one.
- 1.3. Give a second explanation by following what each change does to the current label. Deduce what happens when the first label is 1 and the last is 2, even if the labels change back and forth many times.

A first tool for today's problem

Along a subdivided segment whose labels are only 1, 2, the number of subsegments with different endpoint labels is odd when the two ends of the whole segment have different labels.

Part II — Making the triangle problem precise

We now need a precise meaning for the subdivision and its labelling.

Triangulation and Sperner labelling

Let T be a closed, nondegenerate triangle, meaning that its corners v_1, v_2, v_3 are not collinear. A **triangulation** $\mathcal{T}$ of T is a finite collection of closed, nondegenerate small triangles whose union is T . Two distinct small triangles meet, if at all, in a common vertex or an entire common side. Their interiors do not overlap. The vertices and sides of the small triangles are the **vertices** and **edges of the triangulation**.

A **Sperner labelling** assigns to each triangulation vertex w a label $\ell(w) \in \{1, 2, 3\}$, with:

- $\ell(v_i) = i$ for $i = 1, 2, 3$;
- on the side $[v_i, v_j]$, labels belong to $\{i, j\}$.

There is no additional restriction on interior vertices. A small triangle is **fully labelled**, or a **Sperner triangle**, if its vertex labels are 1, 2, 3, each occurring once.

Closed triangles share their sides; it is their *interiors* that are disjoint. An edge lying on a side of T is a **boundary edge**; all other triangulation edges are **interior edges**.

- 2.1. In the opening figure, call the bottom-left, bottom-right, top and central small triangles a, b, c, d , respectively. List their vertex labels and identify the Sperner triangles.
- 2.2. Explain why the sides of one small triangle cannot stop in the middle of the side of another. Explain also why this definition does not permit one small triangle to contain another.
- 2.3. Justify the following incidence property: a boundary edge belongs to exactly one small triangle, and an interior edge belongs to exactly two. Consider a point strictly between the endpoints of the edge and the two sides of its supporting line.
- 2.4. Call an edge a 12-edge if its endpoint labels are 1 and 2. On which side of T can a boundary 12-edge lie? Use Part I to prove that the number of boundary 12-edges is odd.

Part III — Build a graph that remembers the right edges

Counting fully labelled triangles directly seems difficult. Instead, we record how small triangles meet across 12-edges.

The auxiliary graph $G_{\mathcal{T}}$

Its vertices are the small triangles of $\mathcal{T}$, together with one additional vertex o , representing the outside of T . For every 12-edge:

- if it is interior, join the two adjacent small triangles;
- if it is on the boundary, join its adjacent small triangle to o .

There are no other graph edges. A triangulation edge and an auxiliary graph edge are different objects: the second records the first.

- 3.1. Construct $G_{\mathcal{T}}$ for the opening example, including isolated vertices. Give its edge set and all its degrees.
- 3.2. Verify that the construction gives a finite simple undirected graph. In particular, why can two small triangles contribute at most one edge between them? Why can one small triangle contribute at most one edge to o ?
- 3.3. For a small triangle t , show that its degree in $G_{\mathcal{T}}$ equals the number of its 12-sides. Classify all possibilities for its three vertex labels and prove:

$$\deg(t) \in \{0, 1, 2\}, \quad \deg(t) \text{ is odd} \iff t \text{ is fully labelled.}$$

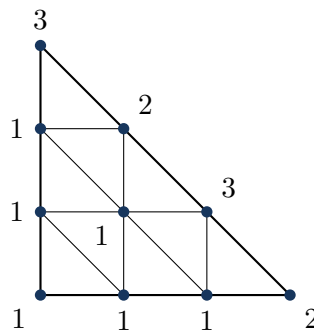
- 3.4. Show that $\deg(o)$ is odd. State explicitly which earlier result supplies this information.
- 3.5. Let N be the number of fully labelled small triangles. Apply the parity theorem from Session 1 to $G_{\mathcal{T}}$. Why is the number of its odd-degree vertices exactly $N + 1$? Deduce both that N is odd and that $N \geq 1$.

Sperner's lemma for a triangle — now proved

Every Sperner-labelled triangulation of a triangle contains an **odd number of fully labelled small triangles**. In particular, at least one exists. The number may depend on the triangulation and the interior labels; its parity does not.

Part IV — What makes the proof work?

- 4.1. Write a short proof of the existence conclusion without referring to the opening figure. Your argument must apply to an arbitrary Sperner-labelled triangulation, not just the four-triangle example.
- 4.2. Remove only the restriction on the labels of vertices strictly between the corners of the big triangle's sides. Construct a triangulation with corner labels 1, 2, 3, but no fully labelled small triangle. What step of our proof is no longer justified? *Hint:* use the four-triangle subdivision and try assigning label 1 to all three side midpoints.
- 4.3. Does the proof establish uniqueness of the fully labelled triangle? Does it give an upper bound independent of the size of the triangulation? Explain what it actually establishes. Check the number of fully labelled triangles in this second example.



- 4.4. The auxiliary graph gives a certificate of existence. Starting at o , explain how following graph edges without reusing an edge leads to a vertex corresponding to a fully labelled triangle. Continue whenever an unused incident edge is available, including after a return to o . Use the fact that all triangle vertices have degree at most two.

Our toolkit after Session 2

- Boundary changes can be counted by induction or by tracking switches.
- An auxiliary graph can turn a geometric existence problem into parity.
- Sperner's lemma gives an odd, hence nonzero, number of fully labelled triangles.
- Boundary assumptions are essential; a counterexample locates their role.

2 Homework — From Sperner's lemma to Brouwer's fixed-point theorem

This work is separate from the three-hour session. Homework A proves the theorem in dimension two. Homework B extends the argument to arbitrary finite dimension, which is needed for probability distributions on an arbitrary finite number of states.

Homework A — A continuous map of a triangle into itself

Consider the standard triangle

$$\Delta^2 = \{x = (x_1, x_2, x_3) \in \mathbb{R}^3 : x_i \geq 0 \ (i = 1, 2, 3), \ x_1 + x_2 + x_3 = 1\}.$$

Its corners are $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$. Although it lies in $\mathbb{R}^3$, this triangle is two-dimensional: it lies in the plane $x_1 + x_2 + x_3 = 1$. Let $f : \Delta^2 \rightarrow \Delta^2$ be a continuous map, written $f(x) = (f_1(x), f_2(x), f_3(x))$.

Fixed point and mesh

A **fixed point** of f is a point x^* for which $f(x^*) = x^*$. For a small triangle t , its **diameter** is $\text{diam}(t) = \sup\{\|x - y\| : x, y \in t\}$, using the Euclidean norm. The **mesh** of a triangulation is the largest diameter of its small triangles.

Our target is the following statement. Do not use it before completing the proof below.

Theorem 1 (Brouwer's fixed-point theorem on a triangle). *Every continuous map $f : \Delta^2 \rightarrow \Delta^2$ has a fixed point.*

- A.1.** Show that Δ^2 is compact. Identify the coordinate inequalities and equality satisfied both by x and by $f(x)$.
- A.2.** Subdivide each small triangle into four by joining the midpoints of its sides. Explain why this gives another triangulation, and why its mesh is half the previous mesh. Starting from the single triangle Δ^2 , construct triangulations $\mathcal{T}_k$ with mesh $h_k = 2^{-k} \text{diam}(\Delta^2)$, for $k \geq 0$.
- A.3.** If a triangulation vertex v satisfies $f(v) = v$, what remains to prove? Otherwise, show that at least one coordinate satisfies $v_i > f_i(v)$. Why is the strict inequality in this direction useful when v lies on the boundary?
- A.4.** Assuming no vertex is a fixed point, label each vertex v with the smallest index i for which $v_i > f_i(v)$. Show that the rule is well-defined and is a Sperner labelling. Check the corners and each boundary side explicitly.
- A.5.** Apply Sperner to $\mathcal{T}_k$. Choose a fully labelled triangle t_k , and denote by $v^{(k,i)}$ its vertex labelled i , for $i = 1, 2, 3$. Write the three strict inequalities provided by their labels. Why are they not necessarily inequalities at the same point?
- A.6.** By compactness, extract a subsequence $v^{(k_j,1)} \rightarrow x^* \in \Delta^2$, where $k_j \rightarrow \infty$. Use $\|v^{(k_j,i)} - v^{(k_j,1)}\| \leq h_{k_j}$ to prove that the other two vertices converge to the *same* x^* .
- A.7.** Use continuity to pass to the limit in the inequalities from A.5. Obtain $x_i^* \geq f_i(x^*)$ for each i . Why can the strict inequalities become weak in the limit?
- A.8.** Since $\sum_i x_i^* = \sum_i f_i(x^*) = 1$, show that all three inequalities must be equalities. Conclude the proof of Theorem 1, including the alternative in A.3.

- A.9.** Suppose T is any closed, nondegenerate triangle with corners a_1, a_2, a_3 . Show that $H(x) = \sum_{i=1}^3 x_i a_i$ is a continuous bijection from Δ^2 to T with continuous inverse. Apply the theorem to $H^{-1} \circ g \circ H$ to deduce the same conclusion for every continuous map $g : T \rightarrow T$.
- A.10.** Locate the two essential hypotheses in the proof. On the interval $[0, 1]$, give a discontinuous self-map with no fixed point, and a continuous map whose image is not contained in $[0, 1]$ and which has no fixed point there. Does the theorem assert uniqueness?

The fixed-point argument in four steps

Label a fine triangulation using $x_i > f_i(x)$; find a fully labelled triangle; let its vertices approach a common limit; use continuity and the coordinate sum to obtain an exact fixed point.

Homework B — The extension needed for more than three states

The theorem just proved concerns a triangle, not a simplex of arbitrary dimension. This extension closes that gap. It is a longer, guided written problem; it need not be presented orally. Use A.3–A.8 after proving the higher-dimensional Sperner lemma.

Simplex, faces and barycentric coordinates

For $d \geq 0$, the **standard d -simplex** is

$$\Delta^d = \{x \in \mathbb{R}^{d+1} : x_i \geq 0 \ (1 \leq i \leq d+1), \sum_{i=1}^{d+1} x_i = 1\}.$$

Its corners are the coordinate vectors $e_1, \dots, e_{d+1}$. More generally, a **d -simplex** with affinely independent corners $a_0, \dots, a_d$ (meaning that the vectors $a_1 - a_0, \dots, a_d - a_0$ are linearly independent) is their convex hull: its points have unique expressions $\sum_{j=0}^d \lambda_j a_j$ with $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$. These coefficients are their **barycentric coordinates**.

A **face** is the convex hull of a nonempty subset of the corners. A **facet** is a face with exactly d corners, of dimension $d-1$, when $d \geq 1$. In Δ^d , the facet opposite e_i is $F_i = \{x \in \Delta^d : x_i = 0\}$. A triangulation is a finite covering by d -simplices whose relative interiors in the affine span of Δ^d are disjoint and whose intersections are common faces. Its mesh is the largest diameter of its simplices.

A **Sperner labelling** uses $1, \dots, d+1$ and satisfies $\ell(v) \in \{i : v_i > 0\}$ at each triangulation vertex. A small d -simplex is **fully labelled** if all $d+1$ labels occur. For $d=2$, these definitions give our triangle problem.

- B.1.** Verify the last sentence of the definition. For $d=0$, the simplex is a single point with the only possible label 1. How many fully labelled small simplices does it contain?
- B.2.** For $d \geq 1$, justify the facet incidence property: each small facet on the boundary belongs to one small d -simplex, and each interior small facet belongs to two. Extend the local two-sided argument of Question 2.3 to the affine span of Δ^d .
- B.3.** Call a small facet *distinguished* when its d vertices carry exactly the labels $1, \dots, d$. Classify the number of distinguished facets of a small d -simplex. Show that it is one in the fully labelled case, two when its labels include all of $1, \dots, d$ but not $d+1$, and zero otherwise.
- B.4.** Explain why a distinguished boundary facet must lie in F_{d+1} . Show that the triangulation and labelling restricted to F_{d+1} form a Sperner-labelled $(d-1)$ -simplex.
- B.5.** Prove by induction on d that every Sperner-labelled triangulation of Δ^d has an odd number of fully labelled small d -simplices. For the induction step, let B and I count distinguished

boundary and interior facets; let N and Q count small simplices with respectively one and two distinguished facets. Count incidences in two ways to obtain

$$N + 2Q = B + 2I.$$

Which side of this identity contains the information supplied by the induction hypothesis?

We also need triangulations of arbitrarily small mesh. For a simplex $S = \text{conv}\{a_0, \dots, a_d\}$, the **barycentre** of a nonempty face F is the average of its corners, denoted $b(F)$. The **barycentric subdivision** uses, for each permutation σ of $\{0, \dots, d\}$, the simplex with vertices

$$b_k = \frac{a_{\sigma(0)} + \dots + a_{\sigma(k)}}{k+1}, \quad 0 \leq k \leq d.$$

These are barycentres of a nested chain of faces. The next questions verify rather than assume that the construction works.

B.6. Let $x = \sum_{j=0}^d \lambda_j a_j \in S$. Choose a permutation with $\lambda_{\sigma(0)} \geq \dots \geq \lambda_{\sigma(d)}$, and put $\lambda_{\sigma(d+1)} = 0$. Set $\mu_k = (k+1)(\lambda_{\sigma(k)} - \lambda_{\sigma(k+1)})$. Show that $\mu_k \geq 0$, $\sum_k \mu_k = 1$ and $x = \sum_k \mu_k b_k$. Deduce coverage. Show that each new simplex is nondegenerate. Explain how equal consecutive coordinates or zero coordinates describe its faces, so different new simplices intersect in common faces and subdivisions agree on shared old faces.

B.7. If $F \subsetneq H$ are two nonempty faces with f and h corners, write

$$b(H) = \frac{f}{h}b(F) + \frac{h-f}{h}b(H \setminus F),$$

where $H \setminus F$ here denotes the face spanned by the corners of H not in F . Deduce $\|b(H) - b(F)\| \leq \frac{d}{d+1} \text{diam}(S)$. Prove that the diameter of a simplex is the maximum distance between its corners, and conclude that repeated barycentric subdivision gives meshes tending to zero for each fixed $d \geq 1$.

B.8. Let $f : \Delta^d \rightarrow \Delta^d$ be continuous. Repeat the labelling and limit argument of A.3–A.8, now with $d+1$ coordinates, using B.5–B.7. Prove the theorem below. Treat $d=0$ separately. Be explicit about why *all* labelled vertices converge along the same subsequence.

Theorem 2 (Brouwer's fixed-point theorem on a finite-dimensional simplex). *For every integer $d \geq 0$, every continuous map $f : \Delta^d \rightarrow \Delta^d$ has a fixed point.*

What becomes available after the homework

Every continuous self-map of the simplex of n nonnegative coordinates summing to one has a fixed point, for any finite $n \geq 1$. The theorem asserts **existence**, not uniqueness or convergence of repeated applications of the map. Those are separate questions for the next session.